

Serhii Kuzmin

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Education

The University of Scranton, Scranton, PA

Bachelor of Science in Computer Science and Mathematical Sciences (Double Major)

May 2026

GPA: 3.86/4.00

- Dean's List: 8/8 semesters
- Honors Society: Upsilon Pi Epsilon, Pi Mu Epsilon

Technical Skills

Languages: Python, Java, Kotlin, C, TypeScript, SQL

Backend Framework & Libraries: FastAPI, Node.js, Bun.js, Python CFFI, REST APIs

Systems & Infrastructure: Git, Docker, Linux, AWS, GitHub/GitLab, GitHub Actions

Databases: PostgreSQL, SQLite, MongoDB

Projects

Surface Evolver Desktop Application

January 2026 – July 2026

- Achieved ~90% backwards compatibility for a legacy mathematical engine by porting it to a cross-platform Tauri desktop app wrapping 50 core engine functions.
- Designed a worker-lifecycle and mutex-serialization layer using a Bun-compiled FFI sidecar process per session, bypassing a single-initialization restriction in the underlying C library.
- Built the RPC dispatch layer for session, file, and persistence operations — including auto-snapshot/restore checkpointing — plus a CI/CD pipeline (CMake, GitHub Actions, CTest/cargo check/tsc) producing Mac, Windows, and Linux installers.

Recipe Diary Android Application

April 2026 – May 2026

- Designed and implemented the local data persistence layer from scratch — SQLite database schema and Room ORM data access layer — supporting recipe creation, management, and offline-first storage.
- Integrated remote data synchronization with an external recipe and meal data provider via a Retrofit-based REST API client, handling network communication between the app and the third-party service.
- Built the application using MVVM architecture in Kotlin with Jetpack Compose, with test coverage across unit, integration, and end-to-end layers validating data integrity and API integration behavior

Full-Stack Group Chat Application

September 2024 – December 2024

- Designed and implemented a full-stack group chat application in C using a client-server architecture.
- Designed and implemented custom communication protocol to speed up the communication by 25% via WebSockets.
- Deployed the server on an AWS-hosted FreeBSD instance with Docker for cross-platform compatibility.

Research Experience

The University of Scranton, Scranton, PA

September 2025 – May 2026

Independent Quantum Computing Student Researcher

- Built and verified hybrid quantum simulation experiments in Python using Qiskit, validating results across 8 distinct experiment configurations.
- Quantified the numerical stability of 1- and 2-qubit Z gates in a Variational Quantum Eigensolver implementation by systematically varying readout error in 5% increments.

Professional Experience

The University of Scranton, Scranton, PA

September 2025 – May 2026

Work Study Assistant

- Promoted Global Education programs through workshops and student engagement.
- Managed budget, spending, and travel requests for the International Students Club.

The University of Scranton, Scranton, PA

October 2023 – December 2025

Peer Tutor

- Provided one-on-one tutoring for Mathematics and Computer Science in Java, and C, covering programming fundamentals, object-oriented design, data structures and algorithms, and core low-level programming principles.
- Improved students' average academic performance by approximately 30% through customized study plans and targeted skill development.
- Facilitated drop-in tutoring sessions for individual students and small groups, addressing Mathematics and Computer Science concepts in real time.